

Malik Tafheem Ul Islam

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Research Interests

Evaluation and reliability of generative and multimodal models: **rule and instruction following in video generation and editing**, **multimodal reasoning**, and **VLM-as-judge evaluation**.

Education

National Institute of Technology, Srinagar

Aug 2025 – Jun 2027

M.Tech in Computer Science and Engineering — **CGPA: 9.18/10** (expected)

- **Key Courses:** Machine Learning (A+), Advanced Algorithms (A+), Introduction to Data Science (A+), Artificial Intelligence and Fuzzy Logic (A+), Computer Vision (A), Network & System Security (A), Real Time Operating Systems (A)
- **In progress (Autumn 2026):** Deep Learning, Advanced Theory of Computation, Big Data, Research Methodology

National University of Sciences and Technology (NUST), Islamabad

Sep 2019 – Jun 2023

BS in Computer Science — **CGPA: 3.64/4.00, rank 16 of 117**

- **Key Courses:** Linear Algebra, Probability and Statistics, Numerical Analysis, Data Structures and Algorithms, Design and Analysis of Algorithms, Artificial Intelligence, Digital Image Processing, Advanced Topics in Computing (computer vision), Theory of Automata, Operating Systems, Computer Networks

Research & Professional Experience

Research Assistant — Video Generative Models & Reasoning

Dec 2025 – Present

GAASH Lab, NIT Srinagar / Srinagar, India

- Investigate **reasoning and instruction following** in video generative models, focusing on temporal, spatial, and causal consistency.
- Build **reproducible inference and validation pipelines** for video generation and editing models.
- Develop **multimodal evaluation workflows** combining automated assessment and human review to examine consistency and instruction adherence beyond visual quality.

Lab Assistant

Sep 2025 – Present

NIT Srinagar / Srinagar, India

- Support the **Python Programming**, **Artificial Intelligence**, and **Network & Security** labs: demonstrate concepts (OOP, NumPy/pandas; search, minimax, CSPs; TCP/IP, packet analysis), debug student programs, and assess submissions.

Research Intern

Dec 2025 – Feb 2026

IIT Roorkee, Department of Electrical Engineering / 6 weeks, supervised by Dr. Abdul Saleem

Mir

- Replicated the **T2I-CompBench++** benchmark end to end: compositional text-to-image evaluation of diffusion models across all 8 sub-categories (attribute binding, spatial and non-spatial relations, numeracy, complex compositions).
- Ran **chain-of-thought and self-consistency** prompting on open LLMs/VLMs in Python and compared the outputs qualitatively against direct prompting.
- Surveyed reasoning in **language, vision-language, and diffusion models**, focusing on reasoning-enhanced generation.

Machine Learning Engineer

Jul 2024 – Jan 2025

Freelance

- Delivered **image classification** solutions using CNNs and linear models, and **clustering** workflows using PCA and K-means.
- Built **TensorFlow and PyTorch** pipelines for preprocessing, model tuning, visualization, and interpretation.

Research Assistant

Apr 2023 – Jun 2024

System Analysis and Verification (SAVe) Lab

- Investigated **image segmentation for surgical robot collision avoidance**; processed surgical datasets to evaluate segmentation methods.
- Implemented **NSGA-II-based image segmentation in Java** for collision avoidance in surgical robots.
- Explored **U-Net, DeepLab, and Mask R-CNN** for segmentation in robotics applications.

Software Engineering Intern

May 2023 – Nov 2023

Bitnine Global Inc.

- Developed **C/C++ utilities for AgentsSQL**, contributing to LDAP synchronization and data loading.
- Configured **ldap2pg**-based synchronization and analyzed **pgloader** data structures to support loader development.

- Reviewed **object detection and tracking** methods, including ByteTrack, FairMOT, YOLO, and EfficientDet.
- Built an annotated **fish dataset** using Supervisely, Roboflow, RectLabel, and LabelImg.
- Applied **transfer learning to YOLO detectors** for fish detection within a tracking workflow.

Machine Learning Intern

May 2021 – Jun 2021

Startup Technologies (Pvt) Ltd.

- Compared **logistic regression and SVM** for ridesharing matching; achieved **82% accuracy** after feature engineering.

Publications & Manuscripts

Intent-Based Networking and Agentic AI for Next-Generation Network Control: A Systematic Survey, Taxonomy, and Research Agenda

Malik Tafheem Ul Islam, Iram Naseer, Mir Salim Ul Islam

Telecommunication Systems (Springer) — Under review (major revision resubmitted, Sep 2026)

A benchmark for rule-based instruction following in video generation and editing

Malik Tafheem Ul Islam*, Iram Naseer*, Tajamul Ashraf* (*equal contribution)

In preparation

Projects

KATHE 2026 Shared Task: English-to-Kashmiri Machine Translation

[GitHub](#) 

- Two-person team (with Iram Naseer): **ranked 7th of 38 teams** on the private test set; fine-tuned **IndicTrans2 (1.1B)** for English to Kashmiri (Perso-Arabic script); public score **13.90** ($\sqrt{\text{BLEU} \times \text{chrF++}}$).
- Fixed a preprocessing error in IndicTrans2 that deleted Arabic short-vowel marks from Kashmiri text. Weights released on Hugging Face.

Anti-Drone System Using Low-Cost Sensors and Machine Learning

Undergraduate Project

- Processed visible-light and infrared drone data; trained **YOLOv5/v7/v8** and **YOLO-NAS** detectors, achieving **~90% mAP**.
- Implemented **SORT** and **DeepSORT** for real-time multi-object tracking.
- Field-tested** smartphone-camera detection under controlled and outdoor conditions.

House Price Prediction

[GitHub](#) 

- Performed **exploratory data analysis and feature engineering** for house-price prediction.
- Achieved **$R^2 = 0.921$ on a 30% test split** with an MLP regressor; built a **Streamlit** prediction interface.

Image Colorization with Conditional GANs

[GitHub](#) 

- Built a **conditional GAN** with a **U-Net/ResNet-18 generator** to predict color channels from grayscale images.
- Trained on approximately **10,000 COCO images**, combining **L1 and adversarial losses** in Lab color space.

Web of Knowledge

[GitHub](#) 

- Built a **C++ bibliographic database** using AVL trees, linked lists, and graphs.
- Implemented **publication and coauthor queries**, including author collaboration checks and graph-distance searches.

Technical Skills

- Research Areas:** Generative AI, video generation and editing, multimodal reasoning, computer vision, model evaluation, agentic AI for network control
- Programming:** Python, C/C++, Java, SQL, R, MATLAB, Assembly
- ML & Data:** PyTorch, TensorFlow, scikit-learn, XGBoost, NumPy, pandas; CNNs, GANs, YOLO, U-Net, DeepLab, Mask R-CNN
- Compute & Models:** HPC with Slurm (H100 MIG), lab GPUs (A10, L4); 12+ open video generation models; VLM-as-judge pipelines (Qwen3-VL); Hugging Face
- Tools:** Git, Docker, LaTeX, Streamlit, Supervisely, Roboflow, RectLabel, LabelImg
- Systems:** Linux (Ubuntu, Kali), Windows, command-line tools

Honors & Achievements

- All India Rank 1388** — GATE Data Science & Artificial Intelligence, 2025
- SAT Subject Tests:** Mathematics **800/800**; Physics and Chemistry **790/800** each
- Academic scholarship** awarded on the basis of GPA at NUST

Certifications

Deep Learning Specialization  (DeepLearning.AI); Machine Learning  (Stanford Online); Associate AI Engineer for Data Scientists  (DataCamp, 40-hour track)